Children’s UX: Usability Issues in Designing for Young People

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Topics:

- Young Users

- Behavior Patterns

Summary: New research with users aged 3–12 shows that children have gained substantial proficiency in using websites and apps since our last studies, though many designs are still not optimized for younger users. Designing for children requires distinct usability approaches, including targeting content narrowly for children of different ages.

Millions of children use the internet, and millions more are coming online each year. Many websites and apps specifically target children with educational or entertainment content, and mainstream organizations often have specific “kids’ corner” sections — either as a public service or to build brand loyalty from an early age.

To **separate design myths from usability facts**, we turn to empirical user research: observations of a broad range of children as they use a wide variety of websites and apps in the United States and other countries.

This research covers **users aged 3–12 years**. (Guidelines for sites targeting people 13–17 years old are available in a report from our separate research with teenagers.)

Usability Studies

We conducted **three separate rounds of usability studies**, testing a total of 125 children (59 girls and 66 boys):

- Study 1 (2001). In this study, we tested **27 sites with 55 children**, aged 6–11. We conducted about a third of the study in Israel, and the rest in the United States.
- Study 2 (2010). In this study, we tested **29 sites with 35 children**, aged 3–12 years. All of these user sessions were in the U.S.
- Study 3 (2018). In this study, we tested **39 sites and 36 apps with 35 children**, aged 3-12 years old. About half of these sessions were conducted in the U.S. and half were in Beijing, China.

In Study 1, we conducted sessions in participants’ homes, at schools, and in a usability lab. All of Study 2 and Study 3 sessions were run in a lab. We tested some users in pairs with a friend or relative, and others individually. Pair sessions worked best for children 6–8 years old. In contrast, for children younger than 6 or older than 8, individual sessions were just as good (and are obviously cheaper, as we had to recruit only one user per session).

Although it can be difficult for shy or very young children, we encouraged users to think out loud while they were using the sites. We told the children that **they were the experts**, and that we wanted them to teach us how children use and think about websites. We then explained that, in order for us to learn, they had to explain what they were thinking at all times.

We mainly took the users to specific sites or apps and gave them **directed tasks** that we'd prepared for each site. Often, these tasks differed according to the user's age and gender. At other times, we gave all users the same task. For example, on 7k7k.com (a Chinese game site), we asked boys and girls to play a ping-pong game. Finally, some sites were tested only by a targeted age group. For example, we only tested the Peppa Pig website and app with 3–5-year-olds, asking them to play a game and watch a video.

We mainly tested websites and apps targeted at children. In Study 1, we also tested a few general websites targeted at grownup users to assess how children use such sites. And, in Studies 2 and 3, we tested some **web-wide tasks**, asking users a general question and letting them find the answer on a site of their choosing. For example, we asked children aged 6–12 to find out how to say "thank you" in Japanese. In Study 3, we included mobile and tablet apps that were popular in the U.S. and China.

In Study 3, we tested 39 websites and 36 apps, covering a **broad range of genres**:

- Education (e.g., ABCmouse, Code Kingdom 编程王国, Funbrain, National Geographic Kids)
- Entertainment (e.g., 123 Draw, Highlights Kids, Roblox, Science vs. Magic 科学大战魔法)
- Government (e.g., Nasa Kids Club, US Mint)
- Media (e.g., Nick Jr., PBS Kids, World of Peppa Pig)
- Museum (e.g., Beijing Museum of Natural History, Idea Museum)
- Toys (e.g., Lego, Barbie)

Changes over Time

We conducted our three studies 8 and 9 years apart (our first children's study was 18 years ago). Most aspects of usability don't change much in that time; when we test the same design questions repeatedly, we usually end up with the same findings. For example, guidelines on the best way to structure a menu or how many items a menu should include are typically the same year after year, because such **user interface questions are determined more by the human brain's characteristics and limitations than by changing technologies**. (Other design issues are more technologically determined. For example, people's approach to watching videos online has changed substantially over the last two decades.)

In addition to the static state of human psychology, **usability findings tend to be stable because the user pool is stable**. If we take a simplified view of the adult audience — say, anyone from 20 to 80 years of age — then 83% of users will be unchanged from one decade to the next. (In reality, most websites don't have an even age distribution among their customers, but the point remains: adult users in 10 years will be generally the same people who are using the Web today.)

Children are a different story, and there are at least three reasons to expect that current usability findings might be different from those of 8 and 17 years ago:

- There has been **high turnover** among the individuals in our 3- to 12-year-old range. Our users from 8 years ago will be between 11 and 20 years old now, so most are no longer within our research's target audience. We're dealing with a completely new generation of kids.
- Between each of our studies, the **amount of time children spend on computers** and mobile devices has increased substantially. And, according to both our research and that of others, the best predictor of how children use websites is how much online practice they have.
- The increased adoption of **mobile devices** has led to kids using the Web and apps at younger ages than previous generations. In the U.S., 98% of homes with children between 0–8 years old had a mobile device in 2017, according to research by Common Sense Media. Within that same age group, 42% of children had their *own* tablet, compared to just 1% in 2011.

Taken together, these observations imply major changes in what constitutes a usable site or app for kids today compared to when mobile devices and computers first entered the home.

Despite this reasonable case for expecting major changes, **our third study actually confirmed most of the guidelines from the previous studies**. For example, in our second study, we observed kids skipping the long paragraphs of instructions on the U.S. Mint Kids site. Eight years later, we saw this same behavior when kids encountered lengthy explanations on the Roblox platform.

Biggest Changes

We did learn many new things, however, and the number of design guidelines increased from 130 to 156 — partly because we looked at new sites that do new things, and partly because we expanded the devices we tested to include mobile devices, tablets, and laptop computers (not just desktop machines).

Compared to our previous study 8 years ago, participants in this study were much **more experienced in using the Internet, although not more experienced with a computer.** Many of our users under 6 had never used a computer before, but they were comfortable browsing apps, games, and websites using tablet and mobile devices.

It was common that our 4 and 5-year-old users already had a year or two of experience interacting with apps and websites, which meant these kids were tech-savvier than the ones we studied 8 years ago. Even 3-year-olds easily recognized the main functions within video players: play, pause, volume controls, and the full-screen icon to make a video bigger. Children swiped horizontally, they scrolled (when the design gave sufficient cues that there was more to see), they used the *Home* button on a tablet to return to the app grid and pick other games. They enjoyed being able to hold a touchscreen and move it comfortably within their view and reach.

This comfort with mobile devices wasn't surprising, given the dramatic increase in mobile device use — globally and especially among children. For example, among children 0–8 years old, Common Sense Media reported that mobile media time (tablets, smartphones, iPods) increased from 5 minutes per day in 2011 to 48 minutes per day in 2017. This generation primarily uses phones and tablets to access online activities from a young age. If they have used a computer, most of them borrowed a parent's laptop. We found that until around age 9, kids in the U.S. and China largely preferred touchscreen devices and trackpads on the PC to a traditional mouse and keyboard.

With this increase in screen time, we observed two main behavior changes since our last study:

- **Little kids, big expectations.** Because children spend more time with devices than 8 years ago, they have more opportunities to get a sense of how things can and should work online. Children expected to be able to tap on and interact with characters and pictures in interfaces, both on touchscreens and desktop displays. (Speaking of touchscreens, several kids tried tapping on nontouchscreen laptop displays and were disappointed when they got no response). Kids expected some sites to play sound or have animation. One pair of 8-year-old girls spent several minutes trying to get a page to play sound and animate, so they could enjoy a game more.

  Despite these higher expectations around interactivity, children did not have the same degree of disappointment that adults did when an app or site didn't work quite as they expected. Yes, children did get frustrated when designs weren't as fast as they liked, but they also were used to a lot of games simply not working, so they shrugged it off as a bug or something beyond their control. Still, given the option, they'd choose to browse or play something else if a site wasn't working well.

- **A willingness to work around difficulties.** There's a myth that children "just know" how to fix a problem or get a website or app to work correctly. That's not generally true (except for a few especially tech-savvy users). However, as kids get experienced with devices, they do become comfortable trying a different approach before giving up entirely. They would refresh the page, close and reopen the browser or app, or use the *Back* button and try again. Though they weren't skilled troubleshooters (they weren't good at problem solving or understanding the root of the issue, and they struggled to interpret error messages), they were very willing to try a few quick solutions. This willingness to experiment is something that users who have grown up with digital devices are more likely to have than users who have entered the digital world later in life. If after a few tries kids still couldn't get something to work, they'd just close the tab, the browser, or tap the device's *Home* button and pick something else do to.

Children vs. Adult Users

The two big conclusions regarding usability for children are:

- **Children and adults are different**, and children need a design style that follows different usability guidelines.
- That said, many of the **things that make sites and apps easier for adults also make them easier for children.** Don't discard what you already know about usable design and how to simplify designs. In particular, comply with UI conventions and employ a consistent design within your site or app.

The following table summarizes some of the main similarities and differences we've observed in user behavior between children (in this study) and adults (across many other studies).

		Children	**Adults**
Same	**Following UI conventions**	Preferred	Preferred
	User control	Preferred	Preferred
	First reactions	Quick to judge site (and to leave if no good)	Quick to judge site (and to le no good)
Small Difference	**Willingness to wait**	Want instant gratification	Limited patience
	Multiple/redundant navigation	Very confusing	Slightly confusing
	***Back* button**	Used in apps and websites when prominent, but browser *back* button not used (young children) Relied on (older children)	Relied on
	Reading	Not at all (youngest children) Tentative (young children) Scanning (older children)	Scanning
	Readability level	Each user's grade level	8^{th} to 10^{th} grade text for broa consumer audiences
	Font size	14 point (young children) 12 point (older children)	12 point (up to 14 point for seniors)
	Scrolling	Avoid (young children) Some (older children)	Some
	Standard gestures on touchscreens (tap, swipe, drag)	Large, simple actions (young kids) Easy and well-liked (older kids)	Easy and well-liked
	Search	Bigger reliance on bookmarks than search, but older children do search	Main entry point to the Web
Big difference	**Goal in visiting websites**	Entertainment	Getting things done Communication/community
	Exploratory behavior	Like to try many options Mine-sweeping the screen	Stick to main path
	Real-life metaphors e.g., spatial navigation	Very helpful for pre-readers	Often distracting or too clunl online UI
	Physical limitations	Slow typists Poor mouse control	None (unless they have disabilities)
	Animation and sound	Liked	Usually disliked
	Advertising and promotions	Can't distinguish from real content	Ads avoided (banner blindne promos viewed skeptically
	Disclosing private info	Usually aware of issues: hesitant to enter info	Often recklessly willing to g out personal info

		Children	Adults
	Age-targeted design	Crucial, with very fine-grained distinctions between age groups	Unimportant for most sites (e to accommodate seniors)

Age-Appropriate Design

A consistent finding in our research over the years is the need to **target very narrow age groups** when designing for children. Indeed, there's no such thing as "designing for children," defined as everybody aged 3–12. At a minimum, you must **distinguish between young (3–5), mid-range (6–8), and older (9–12) children**. Each group has different behaviors, physical and cognitive capabilities and users get substantially more tech-savvy as they get older. And, those different needs range far beyond the obvious imperative to design differently for pre-readers, beginning readers, and moderately skilled readers.

We found that young users reacted negatively to content designed for children that were even one school grade below or above their own level. Children are **acutely aware of age differences**: at one website, a 6-year-old said, *"This website is for babies, maybe 4 or 5 years old. You can tell because of the cartoons and trains."* (Although you might view both 5- and 6-year-olds as "little kids," in the mind of a 6-year-old, the difference between them is vast.)

Finally, it's important to **retain a consistent user experience** rather than bounce users among pages targeting different age groups. In particular, by understanding what attracts children's attention, you can "bury" the links to service content for parents in places that children are unlikely to click. Text-only footers worked well for this purpose.

Advice for Parents and Educators

We conducted this research in order to generate usability guidelines for companies, government agencies, and major non-profit organizations that want to design websites for children. Even so, some of our findings have personal **implications for parents, teachers, and others** who want to help individual children succeed on the internet:

- The main predictor of children's ability to use websites is **their amount of prior experience**. We also found that **children as young as 3 can use websites and apps**, as long as they're designed according to the guidelines for this very young audience. Together, these two findings lead to the advice to start your children on the internet at an early age (while also setting limits; too much computer time isn't good for children).
- Parents and educators should also be aware of how they **model behavior with devices**. Kids learn from what they see around them. In a study by AVG Technologies, conducted with participants in nine countries, 54% of children 8–13 years old felt that parents checked their devices too often. Nearly one third (32%) of **children felt unimportant when their parents were distracted** by mobile device or tablet). Adults must not put the full burden of responsible device use on children, without recognizing the role their own behavior plays in influencing them.

- Campaigns to sensitize children to **the internet's potential dangers** and to teach them to be wary of submitting personal information are meeting with success. Keep up this good work.
- On a more negative note, children still don't understand the web's commercial nature and **lack the skills needed to identify advertising** and treat it differently than real content. We need much stronger efforts to teach children about these facts of new media.

Full Report

The full report on our user research with children, with actionable UX design guidelines is available for download.

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